



University of Wisconsin
**SCHOOL OF MEDICINE
AND PUBLIC HEALTH**

BMI/STAT-768

Statistical Methods for
Medical Image Analysis and
High Dimensional Data

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<https://github.com/laplacebeltrami/BMI768>

Data Complexity

Unit Objectives

- 1) Understand how to measure similarity measures
- 2) Understand correlation can be used to measure nonlinearity
- 3) Understand how to measure geometric similarity

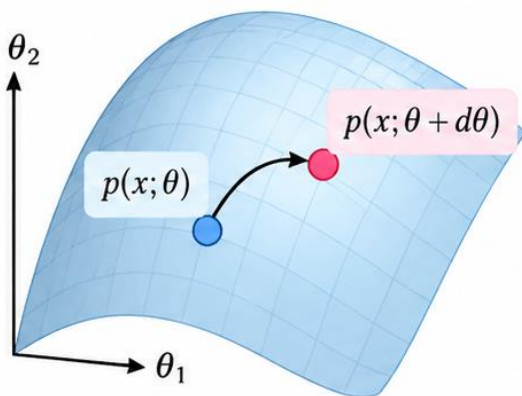
Asymmetric similarity measure connecting to metric tensor

Fisher metric and asymmetric information geometry

Measure distances between probability distributions.

1 Statistical manifold

Each probability distribution $p(x; \theta)$ can be viewed as a point on a smooth manifold parameterized by θ .



Manifold of probability distributions

Nearby distributions (similar θ) lie close on the manifold.

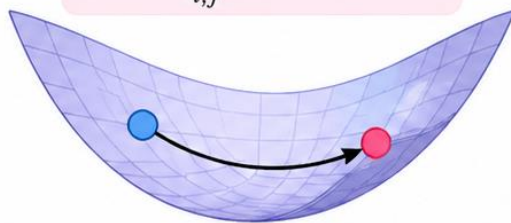
2 Fisher metric (symmetric)

The Fisher information defines a Riemannian metric on the manifold:

$$g_{ij}(\theta) = \mathbb{E} \left[\partial_i \log p(x; \theta) \partial_j \log p(x; \theta) \right]$$

The (squared) distance between two close distributions is

$$ds^2 = \sum_{i,j} g_{ij}(\theta) d\theta^i d\theta^j.$$



Shortest path (geodesic) gives the most efficient change between distributions.

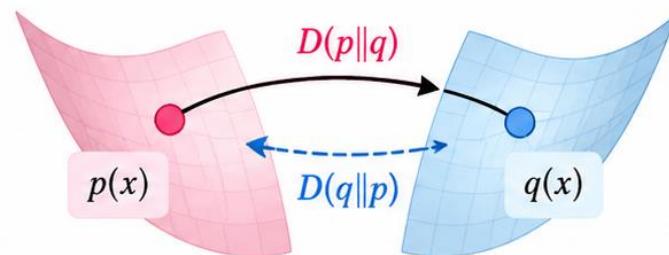
Fisher metric measures local sensitivity of a distribution to parameter changes.

3 Asymmetric (directed) Fisher metric

For directed or time-evolving models (e.g., $p(x_{t+1}|x_t, \theta)$), the Fisher information can be asymmetric:

$$G_{ij} = \mathbb{E} \left[\partial_i \log p(x_{t+1}|x_t; \theta) \partial_j \log p(x_t|x_{t+1}; \theta) \right]$$
$$G \neq G^T$$

This defines an asymmetric distance (or divergence) that captures directional dissimilarity between distributions.



$D(p||q) \neq D(q||p) \rightarrow$ asymmetric information similarity

Useful for modeling directional processes (e.g., temporal dynamics, causal interactions).

Project: Code directed Fisher metric idea surface or time varying network data

What does correlation measure?

It measures linear association, not general dependence.

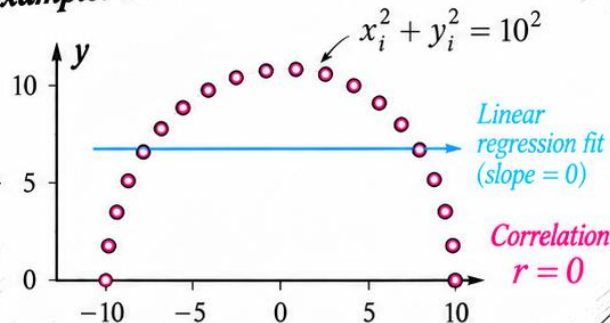
Pearson correlation

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

$$= \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

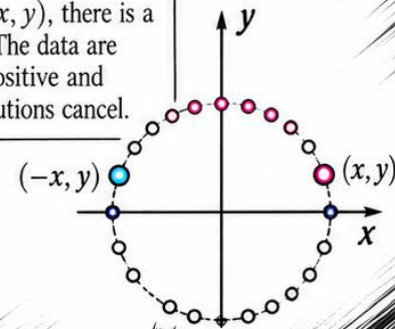
It measures the strength of linear association between X and Y .

Example: a curved relationship



Why is the correlation zero?

For every point (x, y) , there is a point $(-x, y)$. The data are symmetric, so positive and negative contributions cancel.



Mathematical proof

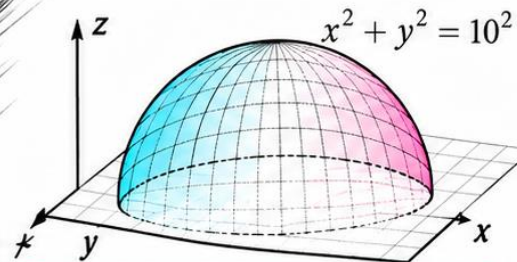
Assume x -values are symmetric and $\bar{X} = 0$.

Since $x_i^2 + y_i^2 = 10^2$, we have $y_i = \sqrt{100 - x_i^2}$, so $y_{-i} = y_i$ for the pair $(-x_i, y_i)$. Then

$$\begin{aligned} \text{cov}(X, Y) &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) \\ &= \frac{1}{n} \sum_{i=1}^n X_i (Y_i - \bar{Y}) \quad (\bar{X} = 0) \\ &= \frac{1}{n} \sum_{i>0} [x_i (y_i - \bar{Y}) + (-x_i)(y_i - \bar{Y})] \\ &= 0. \end{aligned}$$

Hence $r = 0$.

But X and Y are dependent!



Y is a deterministic function of X :

$$Y = \sqrt{100 - X^2}.$$

Nonlinear dependence!

Key takeaway

Zero correlation ($r = 0$) means no linear association, but it does not imply independence.

To detect more general dependence, use higher-order moments or other measures (e.g., mutual information, nonlinear regression).

No correlation $\neq$ No relationship!

Problem: Can correlation be used to measure nonlinearity?

From correlation to independence

Zero correlation does not imply independence (in general).

But if all mixed moments are uncorrelated, then the variables are independent (under mild conditions).

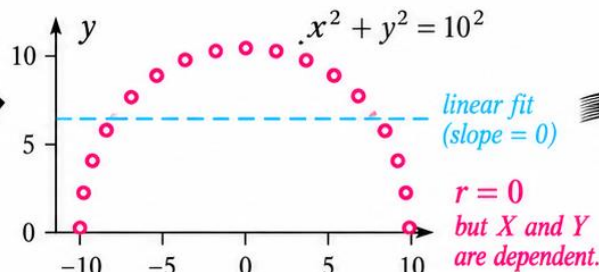


1. Correlation measures **linear** association

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

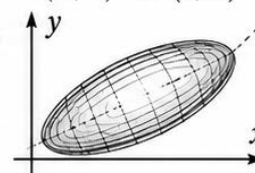
$r = 0$ means no linear relationship, but it does not rule out **nonlinear dependence**.

2. Example: zero correlation but dependent



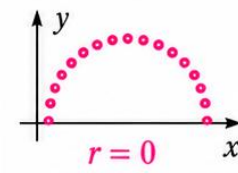
3. Zero correlation $\Rightarrow$ independence only in the **Gaussian** case

$(X, Y) \sim \mathcal{N}(0, \Sigma)$



If (X, Y) is jointly Gaussian, then $\text{corr}(X, Y) = 0 \Leftrightarrow X \perp Y$.

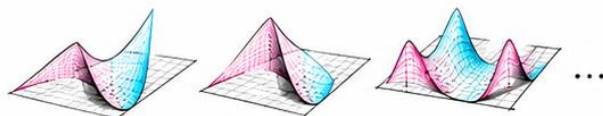
Non-Gaussian case



4. Uncorrelated mixed moments

$$E[X^k Y^j] = E[X^k] E[Y^j]$$

$$\Leftrightarrow \text{Cov}(X^k, Y^j) = 0 \text{ for all } k, j \geq 0.$$



$$E[XY] = 0 \quad E[X^2Y] = 0 \quad E[X^2Y^2] = 0$$

All cross-moments vanish.

5. Why does this imply independence?

If $E[f(X)g(Y)] = E[f(X)]E[g(Y)]$ for all functions f, g (with finite expectations), then X and Y are independent.

$$\text{Cov}(X^k, Y^j) = 0 \quad \forall k, j \geq 0$$

$$\Rightarrow E[f(X)g(Y)] = E[f(X)]E[g(Y)] \text{ for a rich class of functions } f, g.$$

If the joint distribution is determined by its moments (e.g., bounded support, sub-Gaussian, or more generally moment-determinate distributions), then

$$X \perp Y.$$

6. Take-home message

- ❑ $\text{corr}(X, Y) = 0$ does not imply independence (except in the Gaussian case).
- ❑ Zero linear correlation can occur with strong nonlinear dependence.
- ❑ If all mixed moments are uncorrelated ($E[X^k Y^j] = E[X^k]E[Y^j]$ for all $k, j \geq 0$), then X and Y are independent (under mild regularity conditions).

Uncorrelated at all orders $\Rightarrow$ independence.

Problem: Prove the claim.

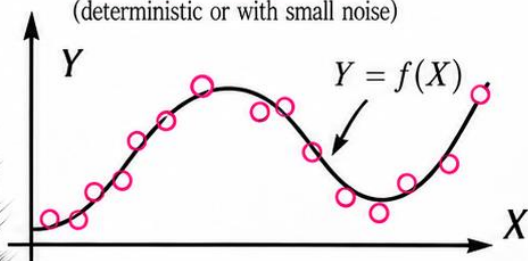
Can we estimate a functional relationship from correlations of moments?

Yes — higher-order correlations (mixed moments) contain the information about the function.

Assume a functional relationship

$$Y = f(X) + \epsilon$$

(deterministic or with small noise)



Mixed moments contain information

$$m_k = E[X^k Y] = E[X^k f(X)]$$

These are inner products of f with monomials.

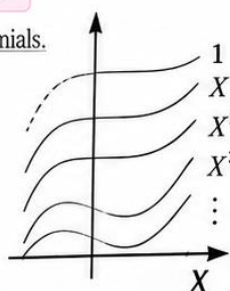
If f can be expanded as a polynomial

$$f(X) = \sum_{\ell=0}^{\infty} a_{\ell} X^{\ell},$$

then

$$E[X^k Y] = \sum_{\ell=0}^{\infty} a_{\ell} E[X^{k+\ell}].$$

Monomial basis



Recover the coefficients from moments

Using the first $(K + 1)$ moments, we solve a linear system

$$\underbrace{\begin{pmatrix} \mu_0 & \mu_1 & \mu_2 & \cdots & \mu_K \\ \mu_1 & \mu_2 & \mu_3 & \cdots & \mu_{K+1} \\ \mu_2 & \mu_3 & \mu_4 & \cdots & \mu_{K+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mu_K & \mu_{K+1} & \mu_{K+2} & \cdots & \mu_{2K} \end{pmatrix}}_{\text{moments of } X} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_K \end{pmatrix} = \underbrace{\begin{pmatrix} m_0 \\ m_1 \\ \vdots \\ m_K \end{pmatrix}}_{\text{mixed moments } m_k = E[X^k Y]}$$

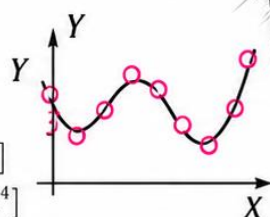
coefficients of f

Example: quadratic relationship

$$Y = aX + bX^2$$

Use $E[XY]$ and $E[X^2Y]$ to solve for a, b .

$$\begin{cases} E[XY] = aE[X^2] + bE[X^3] \\ E[X^2Y] = aE[X^3] + bE[X^4] \end{cases}$$



If the moment matrix is nonsingular (e.g., X not degenerate), we can recover a and b .

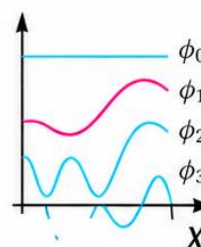
A general and elegant view

Use an orthonormal basis $\{\phi_k(X)\}$ in $L^2(P_X)$, e.g., polynomials (Legendre, Hermite, ...).

$$f(X) = \sum_{k=0}^{\infty} a_k \phi_k(X)$$

$$a_k = E[Y \phi_k(X)]$$

Each coefficient is a (generalized) correlation between Y and a basis function of X .



Important notes

- Ordinary correlation $\text{corr}(X, Y)$ uses only first-order moments and measures linear dependence.
- Higher-order correlations (mixed moments) capture nonlinear dependence.
- If all mixed moments are known and the distribution is moment-determinate (e.g., bounded support, sub-Gaussian), the function f can be uniquely determined.
- In practice, we estimate finitely many moments from data, which gives an approximation of f .
- Correlations alone (standardized moments) are not enough; raw mixed moments (or equivalently covariances) are needed.

Nonlinear structure is encoded in higher-order moments!

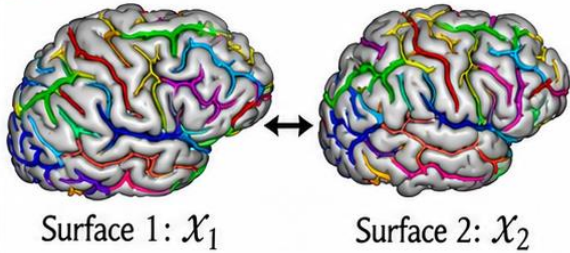
Project: Code underlying idea for various data

Geometric Similarity Measure

How do we measure similarity between brain surfaces?

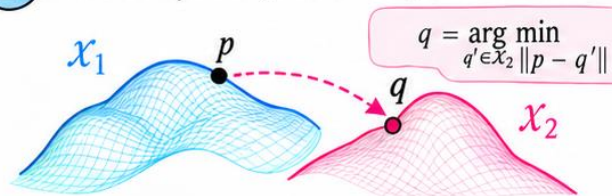
We need a distance (or similarity) to quantify how similar two surfaces are.

1 Brain surfaces to compare



How similar are these two cortical surfaces?
(e.g., left vs. right, across subjects, or across groups)

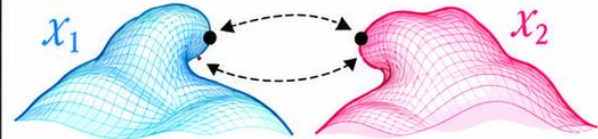
2 Closest-point (one-sided) distance



$$d_{CP}(\mathcal{X}_1, \mathcal{X}_2) = \frac{1}{|\mathcal{X}_1|} \int_{\mathcal{X}_1} \min_{q \in \mathcal{X}_2} \|p - q\|^2 dA(p)$$

Average distance from each point on $\mathcal{X}_1$ to its nearest point on $\mathcal{X}_2$. This distance is not symmetric.

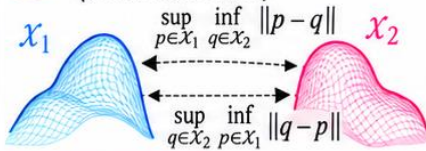
3 Symmetric Chamfer distance



$$d_{Ch}(\mathcal{X}_1, \mathcal{X}_2) = \frac{1}{2} \left(\frac{1}{|\mathcal{X}_1|} \int_{\mathcal{X}_1} \min_{q \in \mathcal{X}_2} \|p - q\|^2 dA + \frac{1}{|\mathcal{X}_2|} \int_{\mathcal{X}_2} \min_{p \in \mathcal{X}_1} \|q - p\|^2 dA \right)$$

Symmetric average nearest-point distance.
Widely used for comparing shapes (point clouds or meshes).

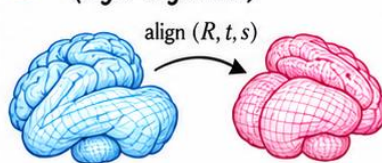
4 Hausdorff distance (worst-case error)



$$d_H(\mathcal{X}_1, \mathcal{X}_2) = \max \left\{ \sup_{p \in \mathcal{X}_1} \inf_{q \in \mathcal{X}_2} \|p - q\|, \sup_{q \in \mathcal{X}_2} \inf_{p \in \mathcal{X}_1} \|q - p\| \right\}$$

Measures the largest mismatch.
Sensitive to outliers, but gives a strong guarantee.

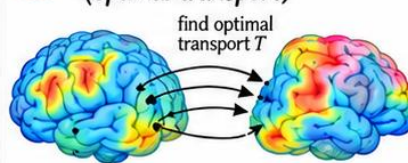
5 Procrustes distance (rigid alignment)



$$d_P(\mathcal{X}_1, \mathcal{X}_2) = \min_{R, t, s} \frac{1}{|\mathcal{X}_1|} \int_{\mathcal{X}_1} \|Rp + t - sq\|^2 dA$$

Remove translation, rotation
(and optionally scale) before comparison.
Good when only shape (not pose) matters.

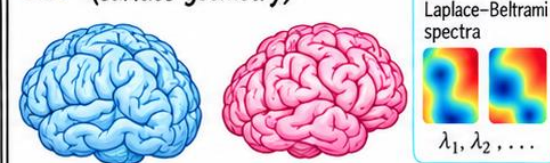
6 Wasserstein distance (optimal transport)



$$d_W(\mathcal{X}_1, \mathcal{X}_2) = \left(\inf_{T \in \Pi} \int \|x - y\|^2 dT(x, y) \right)^{1/2}$$

Finds the optimal one-to-one (mass) transport between two surface distributions.
Captures global shape differences.

7 Spectral / intrinsic distances (surface geometry)



$$d_{spec}(\mathcal{X}_1, \mathcal{X}_2) = \left(\sum_{k=1}^K (\lambda_k^{(1)} - \lambda_k^{(2)})^2 \right)^{1/2}$$

Compare intrinsic geometry via Laplace-Beltrami eigenvalues. Invariant to isometric deformations.

- ✓ Global vs. local similarity?
- ✓ Robust to outliers?
- ✓ Invariant to translation/rotation?
- ✓ Intrinsic geometry or spatial distance?

There is **no single "best" distance** — the choice depends on what kind of **similarity** matters for your application.

Project: Code underlying idea for surface data, and develop a new inference procedure

Information-theoretic similarity measures

Measure similarity by how much **information** is shared.

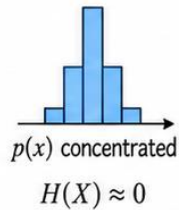
1 Entropy

Amount of uncertainty in a single variable

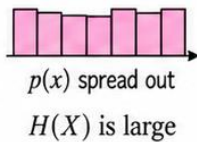
Entropy measures the average uncertainty (or randomness) of a random variable.

$$H(X) = - \sum_i p(x_i) \log p(x_i)$$

Low entropy
(more certain)



High entropy
(more uncertain)



Entropy tells us how unpredictable a variable is.

Larger shared information means more similar; larger divergence means more different!

2 Mutual information

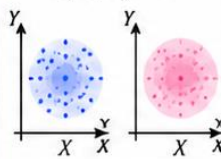
Amount of shared information

Mutual information measures how much knowing one variable reduces the uncertainty of the other.

$$I(X; Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

Independent

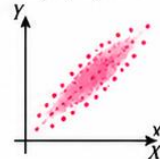
$$I(X; Y) = 0$$



Knowing X tells us nothing about Y .

Dependent

$$I(X; Y) > 0$$



Knowing X reduces uncertainty about Y .

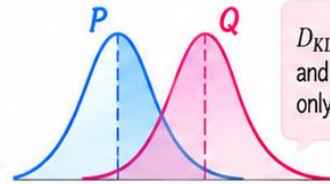
Larger $I(X; Y)$ means the two variables are more similar (more shared information).

3 Kullback-Leibler divergence

How different two distributions are

KL divergence measures how much information is lost when we use Q to approximate P .

$$D_{KL}(P \parallel Q) = \sum_i p(x_i) \log \frac{p(x_i)}{q(x_i)}$$



$D_{KL}(P \parallel Q) \geq 0$
and equals 0 only if $P = Q$.

- Measures the dissimilarity between two probability distributions.
- Not symmetric: $D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$.
- Widely used in statistics, machine learning, and image analysis.

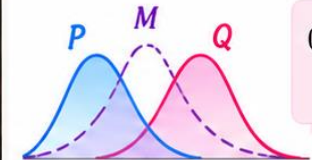
4 Jensen-Shannon divergence

A symmetric similarity measure

Jensen-Shannon divergence is a symmetric and bounded version of KL divergence. It can be converted to a similarity measure.

$$JS(P, Q) = \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M)$$

$$M = \frac{1}{2}(P + Q)$$



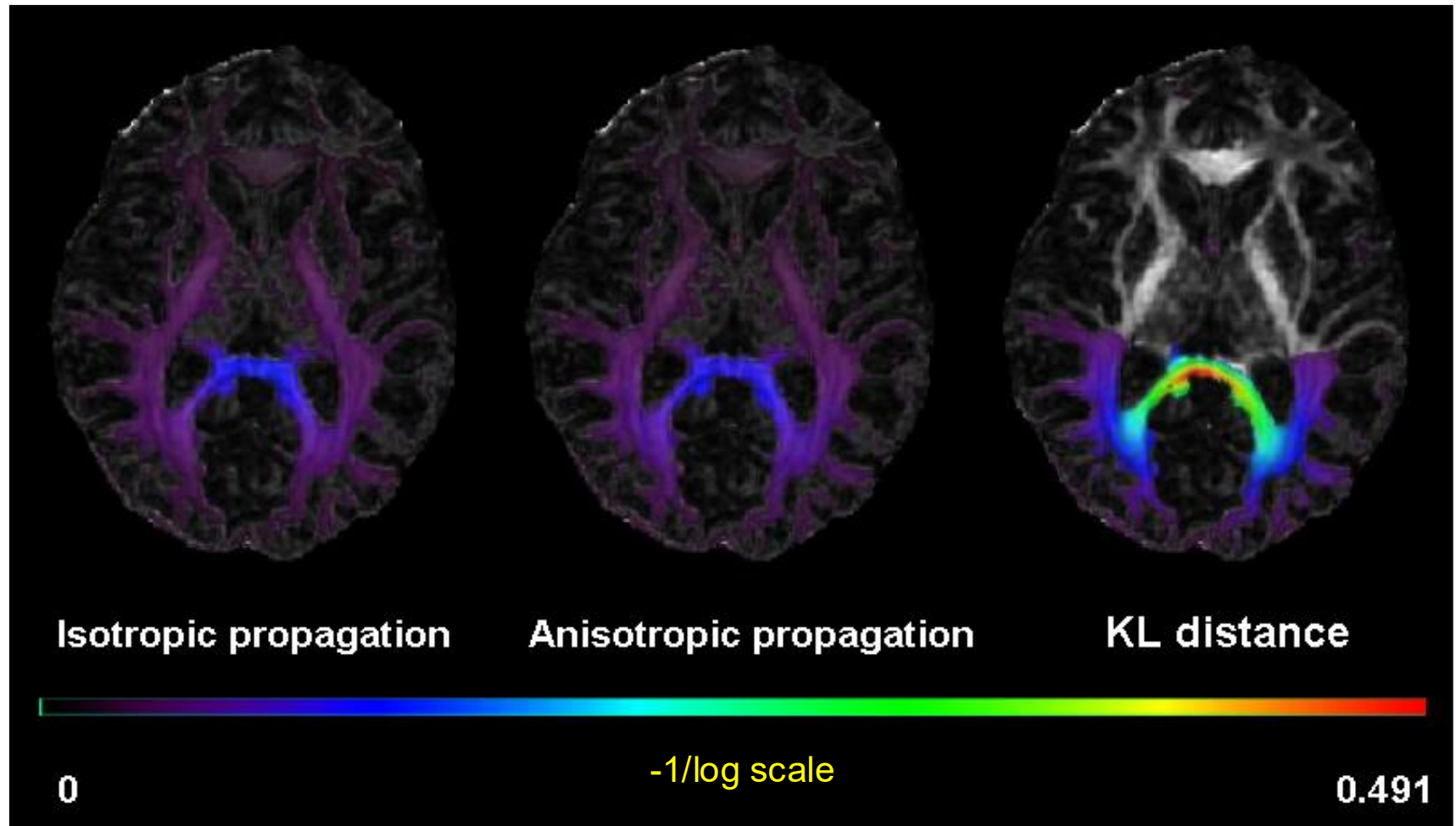
$$0 \leq JS(P, Q) \leq \log 2$$

- Symmetric: $JS(P, Q) = JS(Q, P)$.
- Bounded: 0 to $\log 2$.
- Often used to define a similarity measure, e.g., $S(P, Q) = 1 - \frac{JS(P, Q)}{\log 2}$.

Same framework: compare objects by their probability distributions and use **information-theoretic measures**.

These ideas can be applied to many data types: brain surfaces, images, signals, shapes, etc.!

Diffusion tensor imaging connectivity



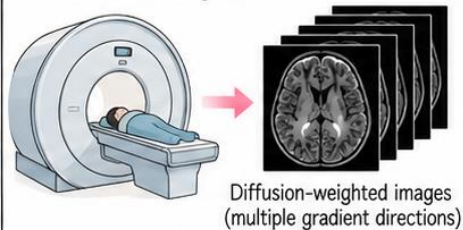
KL-distance is computed within a small window around each voxel

DTI connectivity via KL-distance

From diffusion tensors to probability distributions, then compare with KL divergence.

1 Diffusion Tensor Imaging (DTI)

DTI measures the diffusion of water molecules in the brain using diffusion-weighted MRI.



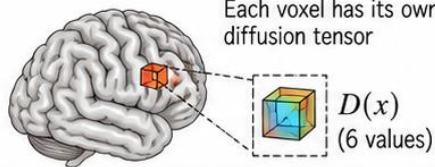
From these images, each voxel x is modeled by a 3×3 diffusion tensor (a symmetric matrix):

$$D(x) = \begin{pmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{xy} & D_{yy} & D_{yz} \\ D_{xz} & D_{yz} & D_{zz} \end{pmatrix}$$

Thus DTI gives a 4D image:

- 3 spatial dimensions (voxels)
- 6 independent tensor components at each voxel (3×3 symmetric)

Each voxel has its own diffusion tensor



2 From tensor to probability distribution

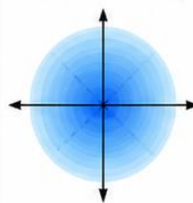
Under the Gaussian diffusion model, the displacement r of a water molecule over time t at voxel x has probability density

$$p_D(r | x) = \frac{1}{(4\pi t)^{3/2} |D(x)|^{1/2}} \exp\left(-\frac{r^T D(x)^{-1} r}{4t}\right)$$

$r \in \mathbb{R}^3$

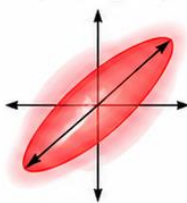
So each voxel defines a 3D Gaussian distribution with covariance proportional to $D(x)$.

Isotropic diffusion
($\lambda_1 \approx \lambda_2 \approx \lambda_3$)



Equal probability in all directions.

Anisotropic diffusion
($\lambda_1 \gg \lambda_2, \lambda_3$)



Higher probability along the fiber direction.

3 Isotropic reference and KL divergence

To measure how directional the diffusion is, we compare the voxel's distribution with an isotropic reference distribution that has the same overall diffusivity:

$$D_{iso}(x) = \bar{\lambda}(x)I, \quad \bar{\lambda}(x) = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3}$$

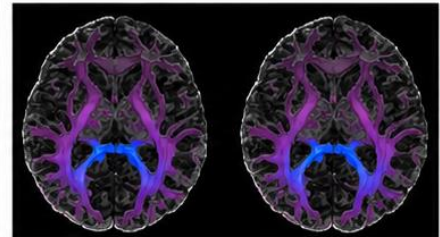
The difference is quantified by the Kullback–Leibler (KL) divergence:

$$D_{KL}(x) = \int p_D(r | x) \log \frac{p_D(r | x)}{p_{iso}(r | x)} dr$$

- $D_{KL}(x) \geq 0$, and equals 0 if the diffusion is isotropic.
- Larger $D_{KL}(x)$ means stronger anisotropy (more directional diffusion).
- KL is computed locally, using a small neighborhood around each voxel to estimate stable tensors.

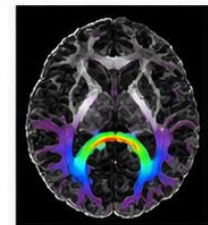
4 Voxelwise KL map

Compute $D_{KL}(x)$ within a small window around each voxel to obtain a KL map that highlights coherent white matter structures.



Isotropic propagation (reference)

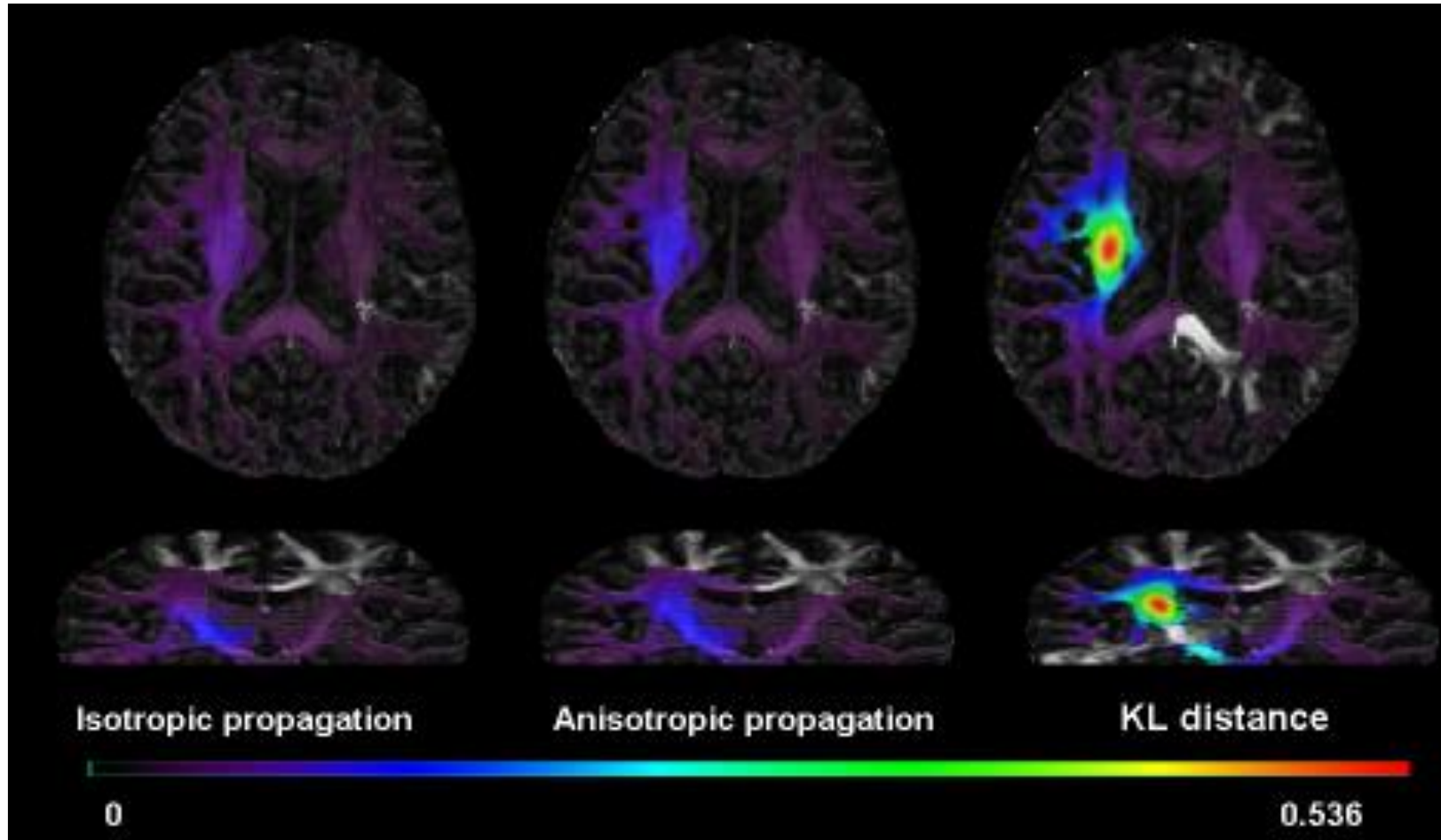
Anisotropic propagation (from DTI)



KL distance



Statistical inference on KL-distance



Null model

Alternate model

Significance of connectivity can be also tested.

Asymmetric similarity measure

Transfer Entropy

How much does the past of one region help predict the future of another region?

A directed, model-free measure of information flow.

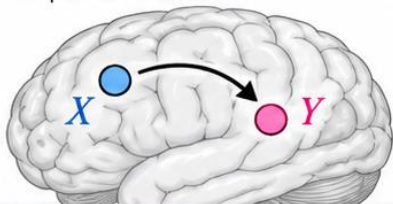
1 Time series from two brain regions

BOLD (or neural) signals from region X and region Y



We ask:

Does the past of X help predict the future of Y , beyond the past of Y itself?



2 Definition

Transfer entropy from X to Y is the conditional mutual information

$$T_{X \rightarrow Y} = I(Y_{t+1}; X_t^{(k)} | Y_t^{(l)})$$

It measures how much knowing the past of X ($X_t^{(k)} = [X_t, X_{t-1}, \dots, X_{t-k+1}]$) reduces the uncertainty of Y_{t+1} , given the past of Y ($Y_t^{(l)} = [Y_t, Y_{t-1}, \dots, Y_{t-l+1}]$).

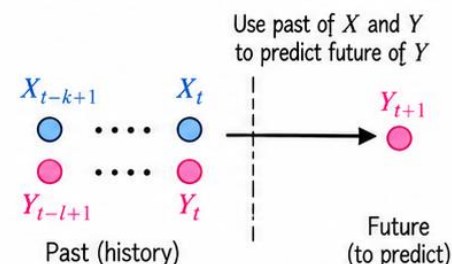
$$T_{X \rightarrow Y} = H(Y_{t+1} | Y_t^{(l)}) - H(Y_{t+1} | Y_t^{(l)}, X_t^{(k)})$$

$H(\cdot)$: conditional entropy

$I(\cdot; \cdot | \cdot)$: conditional mutual information

3 How to compute (practically)

- 1 Choose embedding lengths k and l (e.g., via cross-validation).
- 2 Reconstruct state vectors $X_t^{(k)}, Y_t^{(l)}$.
- 3 Estimate probability distributions (e.g., k -nearest neighbors or binning).
- 4 Compute conditional entropies $\rightarrow T_{X \rightarrow Y}$.



4 Interpretation

$$T_{X \rightarrow Y} > 0$$



X provides predictive information about future Y (directed influence).

$$T_{X \rightarrow Y} \approx 0$$



Past of X does not help predict future Y (no directed influence).

$$T_{X \rightarrow Y} < T_{Y \rightarrow X}$$



Asymmetric information flow.

Key point:

Transfer entropy detects directed, possibly nonlinear dependencies without assuming a specific model (e.g., not limited to linear Gaussian).

Project: Transfer entropy cannot yet to be applied to cyclic interactions. How we modify the information similarity for cyclic interactions?

Unit outcomes

- 1) Understand how to measure similarity measures
- 2) Understand correlation can be used to measure nonlinearity
- 3) Understand how to measure geometric similarity